**Experiment No. 7 (Software)** **Date: 12.11.2022**

**Counters**

Aim:

To design and verify up-down, up, down counters using Verilog Programming.

**Software Required:**

ModelSim

**Procedure:**

1. Open ModelSim.
2. Create New Project, then create a new file by selecting the verilog in the drop-down.
3. Type the code, then compile it.
4. Type the test bench, compile and simulate it.
5. Click add wave in the window and then run it.

**Program Code:**

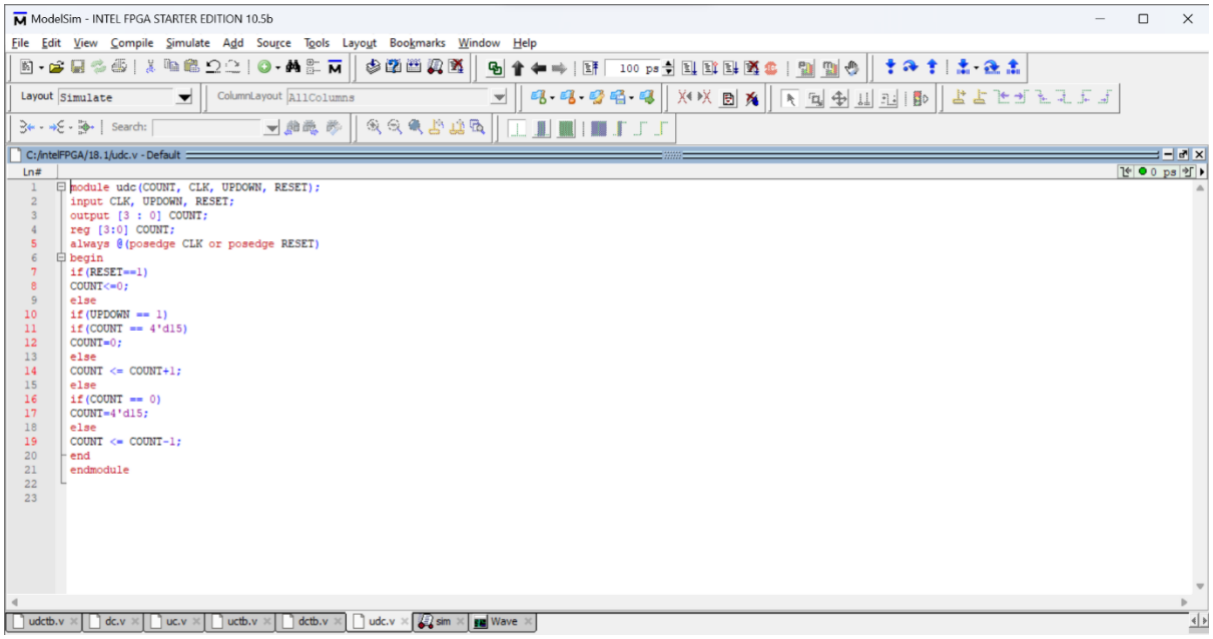
*Up-Down Counter:*

```
module udc(COUNT, CLK, UPDOWN, RESET);

input CLK, UPDOWN, RESET;

output [3 : 0] COUNT;

reg [3:0] COUNT;

always @(posedge CLK or posedge RESET)

begin

if(RESET==1)

COUNT<=0;

else

if(UPDOWN == 1)     //Up mode selected

if(COUNT == 4'd15)

COUNT=0;

else

COUNT <= COUNT+1;

else

if(COUNT == 0)

COUNT=4'd15;

else
```

COUNT <= COUNT-1;

end

endmodule




*<u>Test Bench:</u>*

module udctb;

reg CLK, UPDOWN, RESET;

wire [3 : 0] COUNT;

udc uut(COUNT, CLK, UPDOWN, RESET);

always  #5 CLK=~CLK;

initial

begin

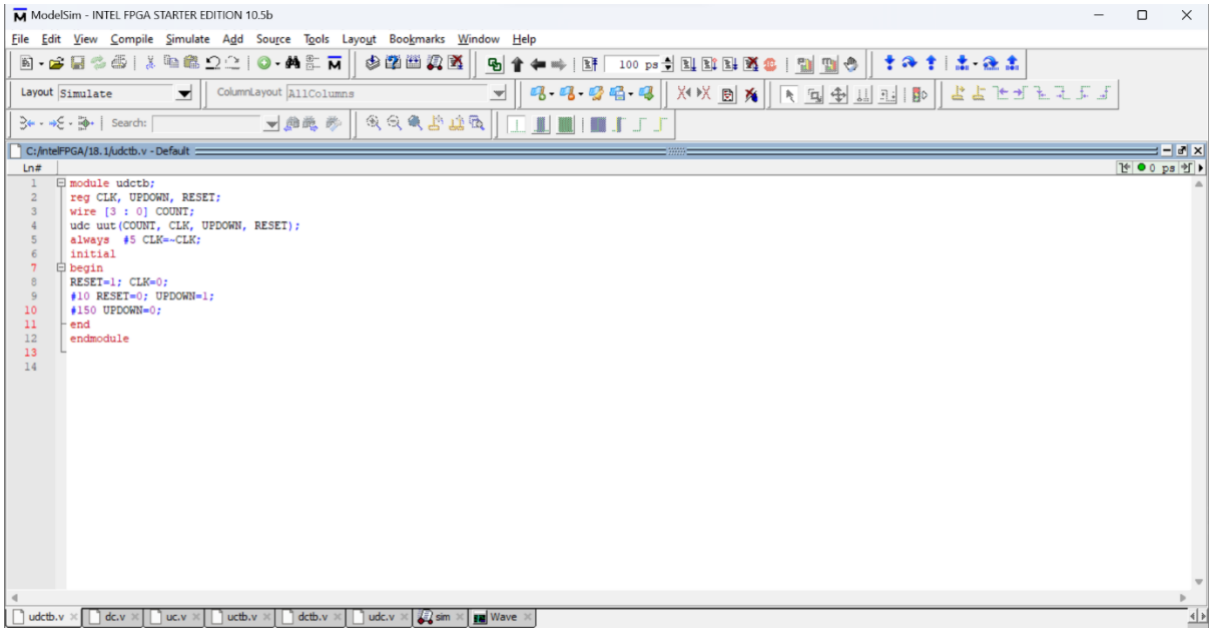
RESET=1; CLK=0;

#10 RESET=0; UPDOWN=1;

#150 UPDOWN=0;

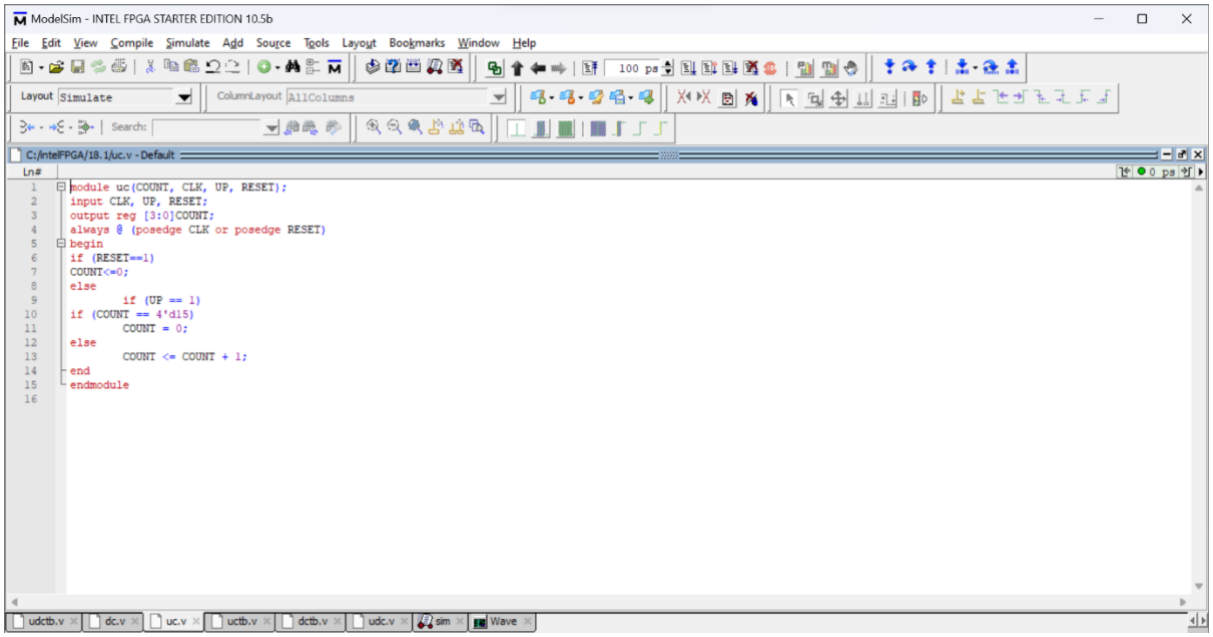
end

endmodule



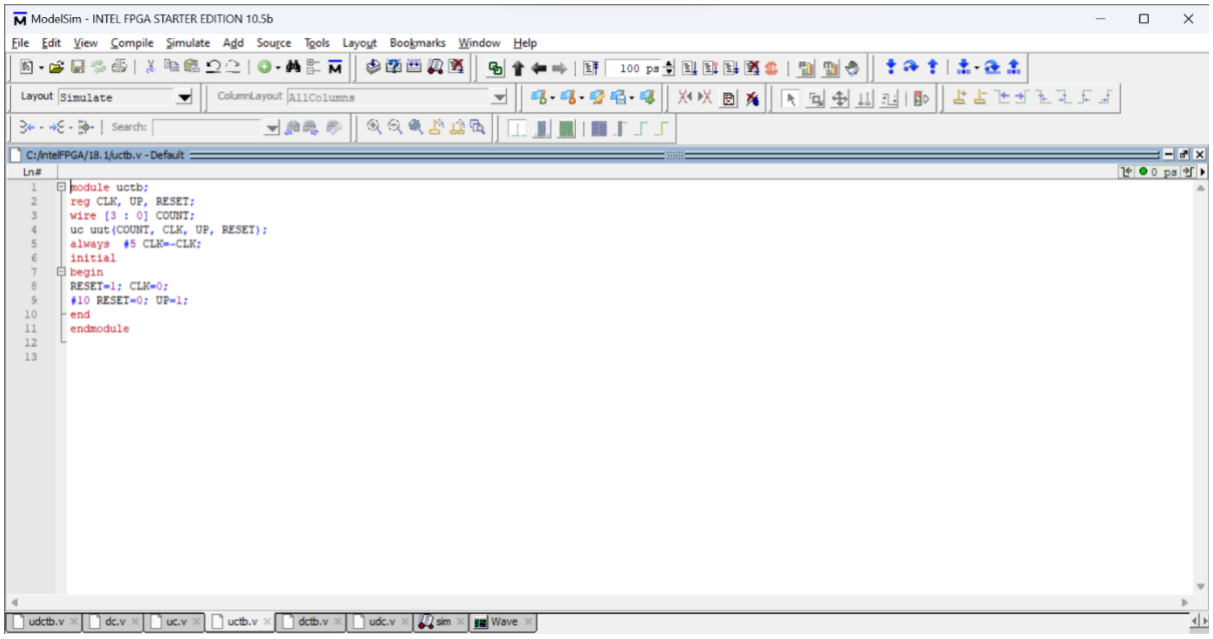

*<u>Up Counter:</u>*

```
module uc(COUNT, CLK, UP, RESET);
input CLK, UP, RESET;
output [3 : 0] COUNT;
reg [3:0] COUNT;
always @(posedge CLK or posedge RESET)
begin
if(RESET==1)
COUNT<=0;
else
        if(UP == 1)
if(COUNT == 4'd15)
        COUNT=0;
else
COUNT <= COUNT+1;
end
endmodule
```



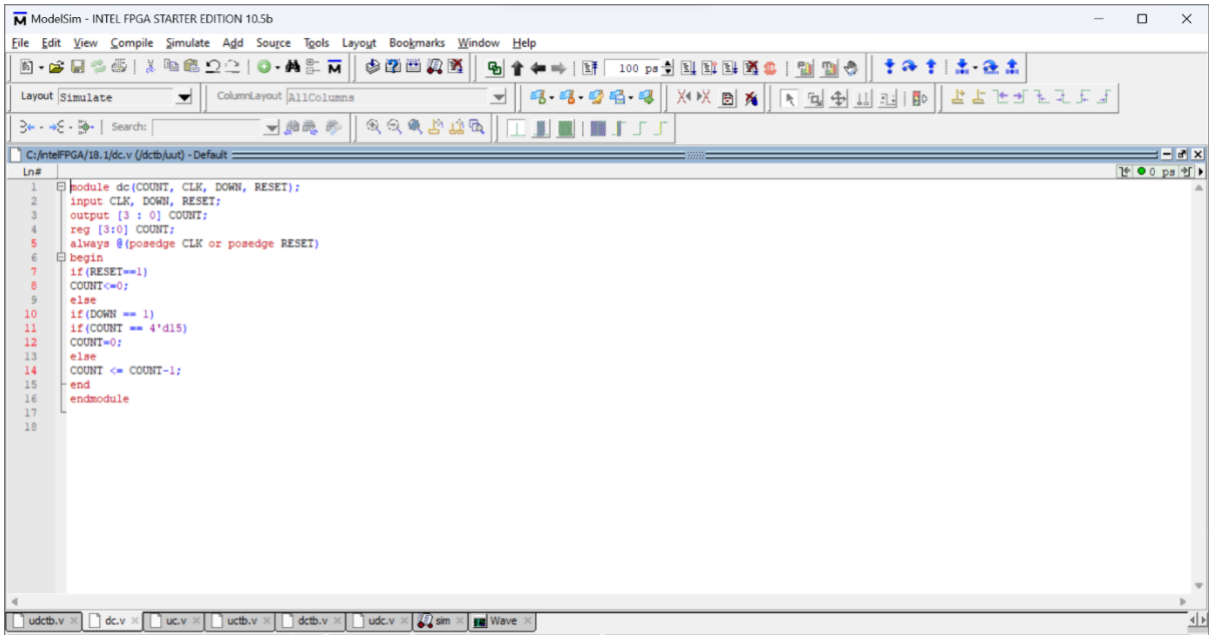

*Test Bench:*

```
module uctb;
reg CLK, UP, RESET;
wire [3 : 0] COUNT;
uc uut(COUNT, CLK, UP, RESET);
always  #5 CLK=~CLK;
initial
begin
RESET=1; CLK=0;
#10 RESET=0; UP=1;
end
endmodule
```



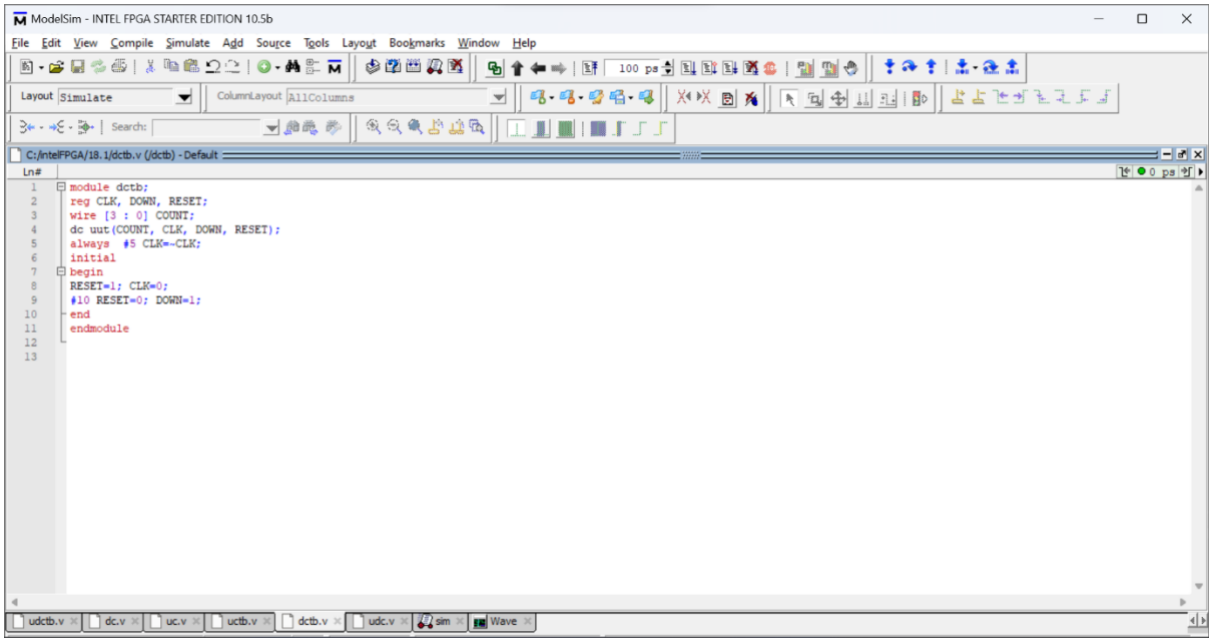

*Down Counter:*

```
module dc(COUNT, CLK, DOWN, RESET);
input CLK, DOWN, RESET;
output [3 : 0] COUNT;
reg [3:0] COUNT;
always @(posedge CLK or posedge RESET)
begin
if(RESET==1)
COUNT<=0;
else
if(DOWN == 1)
if(COUNT == 4'd15)
COUNT=0;
else
COUNT <= COUNT-1;
end
endmodule
```



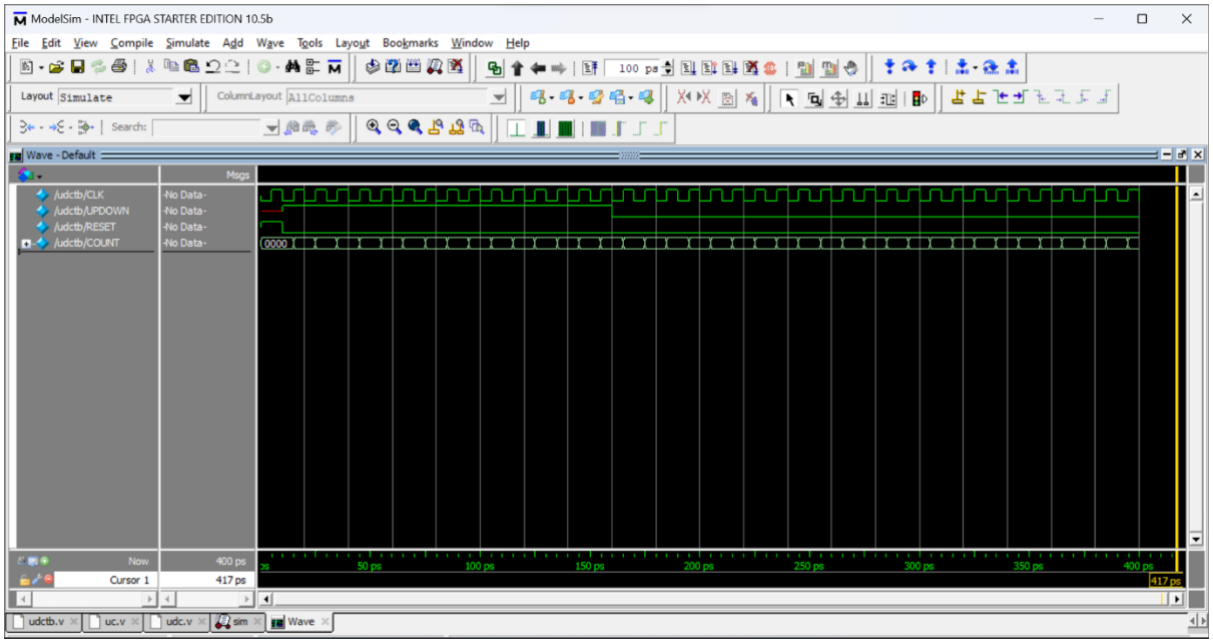

*Test Bench:*

```
module dctb;
reg CLK, DOWN, RESET;
wire [3 : 0] COUNT;
dc uut(COUNT, CLK, DOWN, RESET);
always  #5 CLK=~CLK;
initial
begin
RESET=1; CLK=0;
#10 RESET=0; DOWN=1;
end
endmodule
```

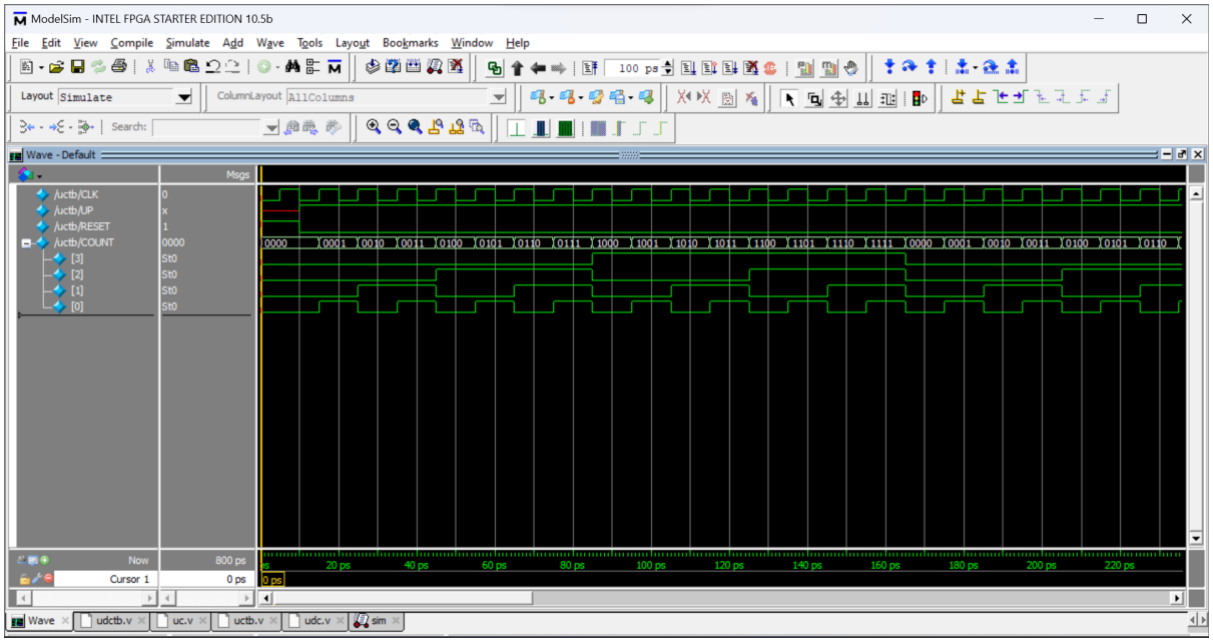



**Output:**

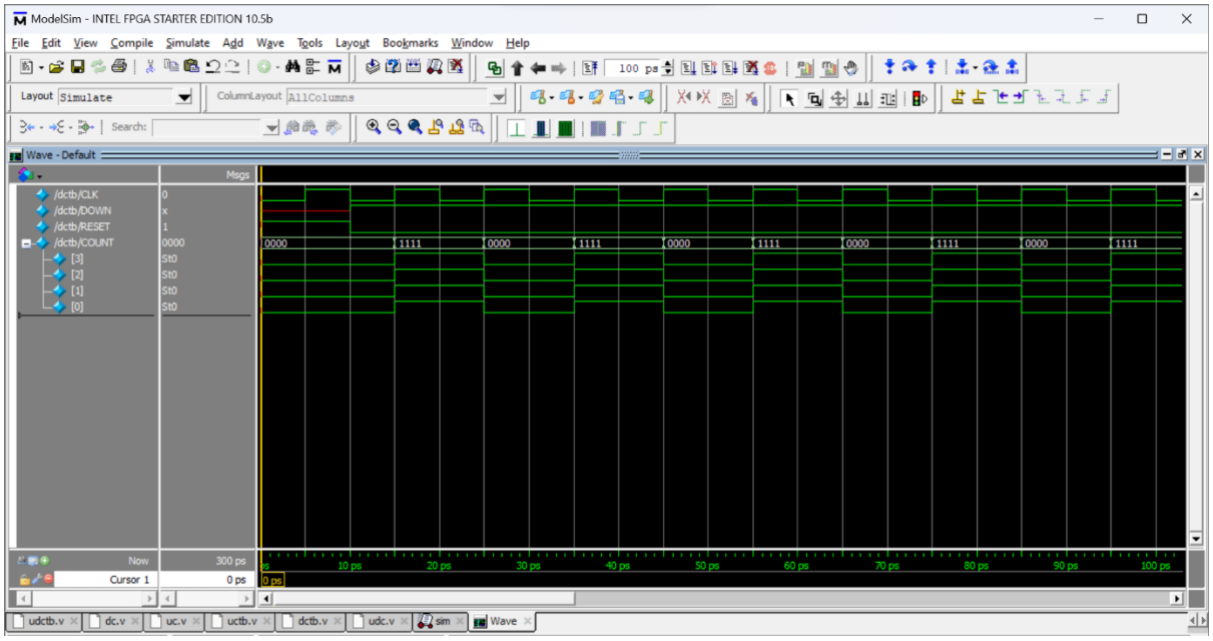
*<u>Up-Down Counter:</u>*




*<u>Up Counter:</u>*




*Down Counter:*




**Result/ Inference:**

The up-down, up and down counters were designed and verified using Verilog Programming.